

The Ultimate Interface

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ABSTRACT

This talk will explore the possibilities of what emerging AI architectures could mean for interaction. My perspective and examples are based on personal opinions, informed by observations and experience gained from building advanced interactions over the last few decades. While I aim to provide insights, speculations, and suggest some essential conditions for creating such an ultimate interface, my insights may not be exhaustive. I invite open discussion and debate on this subject, recognizing that the conditions that I propose might be necessary but perhaps not sufficient in themselves.

CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms.**

KEYWORDS

Futurism, Conceptual, Interaction, AI

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1 INTRODUCTION

In 2014, I joined Google and initiated a project that would later be known as Soli [3].

The goal was to develop the ultimate sensor for ubiquitous and wearable computing: a tiny radar capable of capturing the intricacies of human behavior. This radar would ensure user privacy and be small enough to integrate into a smartwatch. Interfaces built with Soli would understand user intent and respond even before an explicit request was formulated, serving as the ultimate interface for future spatial computers.

At the time I started working on Soli, voice input was gaining traction. With the launch of Amazon Alexa, I was frequently asked, "Why do you need gestures and body capture at all? Voice input will solve everything; voice is the ultimate interface. Just like in Star Trek."

We successfully built Soli and released it into the world through Google products, making it available to consumers in several countries. Although the adoption of general-purpose radar inspired by Soli is growing, it's taking longer than I anticipated. Smart speakers and general-purpose speech input did not become the ultimate

interfaces [2]. Similarly, spatial computing has yet to establish itself as the ultimate interface and is primarily used in niche areas like VR. For most consumers, the ultimate interface remains the touchscreen and graphical user interface (GUI), concepts that have remained largely unchanged since the 1980s.

But recently, a series of transformative technologies have emerged. Among the most profound is the development of transformer-based Large Language Models (LLMs) that can generate extremely rich and nuanced textual representations of human concepts and the relationships between them. These developments have once again raised a fundamental question: Can we move beyond the "point-and-click" interaction model? Is it possible to create the ultimate interface that isn't based on screen-based interaction?

This talk will explore the implications of emerging AI architectures for interaction in general, and specifically, the challenges they could address to make new interaction models a reality. My perspective and examples are rooted in personal opinions, shaped by observations and experiences from building advanced interactions over the past few decades. While I aim to offer insights, make speculations, and suggest some essential conditions for crafting the ultimate interface, I acknowledge that my insights might not be exhaustive. I welcome open discussion and debate on this topic, recognizing that the conditions I propose may be necessary but not necessarily sufficient.

2 TRUE MULTIMODALITY

The first condition for creating an ultimate interface is achieving true multimodality that scales. In the realm of user interaction, the traditional understanding of multimodality is that user intent is captured by combining multiple modes of interaction, such as gestures and voice, as first described in the "Put That There" paper [1]. Despite significant research and development efforts, the adoption of multimodality in real-world applications has been modest at best. In my opinion, this modest adoption is because sensor fusion at the data level is technically rather challenging. As a result, the fusion of modalities often occurs at the heuristic layer, where interaction designers make handcrafted decisions about how modalities should be combined.

This approach is not scalable, and it doesn't align with how humans process information. We don't have separate audio, visual, or tactile streams for each sensory channel that we analyze in isolation to make sense of our surroundings. Instead, our perceptual systems fuses all sensor inputs into a unified intermediate representation that can be accessed by higher cognitive mechanisms to make cross-modal decisions. To realize true multimodal interaction, we need a single, *unified representation* of all sensor inputs, such as voice captured by microphones, gestures and body postures detected by cameras, heartbeat and breathing rates measured by radar, lidar 3D scans, signals from brain implants, gaze direction, and more.

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This "Sensor Rosetta Stone" would allow us to understand the full spectrum of user intent, both conscious and subconscious.

Could this be achieved using multimodal LLMs? One possibility is to convert all sensor modalities into a shared embedding space that is then fed into a transformer model to interpret user intentions. This approach could bring us closer to a true multimodal interface, where the fusion of different modalities occurs seamlessly.

3 CONTEXT THAT WORKS

The second critical challenge is integrating context on a large scale. The importance of context is well-recognized in HCI. Indeed, the same action, such as a person waving their hand, can take on different meanings depending on the physical context – whether it's during a lecture to ask a question or at the airport to greet someone. The interface should discern these contextual differences and generate appropriate responses. However, introducing relevant context into real-world use cases is intricate and challenging. While limited context can be deduced from individual signals, manually crafting it for each application is neither scalable nor easy to extend beyond simple straightforward use cases.

Can LLMs be employed for automatic context generation? By merging sensor data into the embedding space of the LLMs, we might harness the knowledge contained within the LLM to produce a semantic model of the immediate environment, deriving spatial, temporal, and social contexts to clarify the interaction. Spatial context could be autonomously generated from location data and observation of local objects. Social context could arise from detecting the presence of others and discerning cultural norms and social roles through LLMs. Similarly, temporal context can be produced; for instance, the same action in the morning and evening might be interpreted differently.

4 GENERATIVE UI

Do we even need to design interfaces anymore? I would contend that if we manage to seamlessly integrate multimodality and context within a singular large model, the entire interaction might be generative, evolving in real-time. Traditional user interface design could become a relic of the past, supplanted by interfaces that adeptly predict user needs and display information in ways that best align with both the user and their immediate environment, be it on a phone, through a loudspeaker, or via a humanoid robot if one is available nearby.

To achieve this hyper-scaled interface, everything should be connected to and streamed into the ultimate interface. Edge computing would capture only the low-level sensor data, pre-process it, and then stream it to the cloud. This approach should also ensure the preservation of privacy and security.

5 ONE MODEL

All of the above cannot be constructed piecemeal as individual subsystems. They must be built as a single, large AI model that is multimodal, real-time, generative, and contextual both in time and space. Let's term this speculative, yet-to-be-developed model the "Large Behavior Model" or LBM. It's an artificial intelligence that comprehends the entire complexity of human behaviors in relation to the behavior of local environments and machines. This model

deduces the meaning of human intent and generates a response dynamically, based on a holistic understanding of what people need and want.

Such artificial intelligence would be an ultimate interface.

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